

Artificial Intelligence Midterm Exam

Extracted Questions Only

PART-1 (20 Marks)

Note: Choose the best option for the following questions.

1. For a graph search problem, if the cost of every action (move) is the same, breadth first search is guaranteed to return an optimal solution.
(A) True (B) False
2. For a 3×3 tic-tac-toe game, there are _____ different paths (from start to the end of game) in its state space graph.
(A) 3^9 (B) 9^3 (C) $8!$ (D) None of the given options
3. If symmetry of initial symbol placement is considered, the above tic-tac-toe game has _____ paths through its state space graph.
(A) $3 \times 7!$ (B) $12 \times 7!$ (C) $8 \times 7!$ (D) None of the given options
4. Which of the following is true about data-driven search?
(A) It is also called backward chaining
(B) It starts at the goal node(s)
(C) It involves searching from facts to the goal(s)
(D) It is feasible when goal(s) of a problem are known.
5. For a shallower search tree, depth first iterative deepening search compared with depth first search, in general, is:
(A) More efficient (B) Equally well (C) Optimal (D) Inefficient
6. For the 8-puzzle problem, the state space has _____ states accessible from any random start state.
(A) $8!$ (B) $9!$ (C) $9!/2$ (D) None of the given options
7. $\alpha\beta$ search is always better than the minimax search.
(A) True (B) False
8. α cut-off means:
(A) Prune further possibilities below the current MAX node
(B) Prune further possibilities below the current MIN node
9. If d represents depth and b , the average branching factor, the worst-case space complexity of $\alpha\beta$ search can be characterized as:
(A) $O(b^d)$ (B) $O(b \log d)$ (C) $O(d^b)$ (D) $O(bd)$
10. Admissibility of a heuristic always implies:
(A) Informedness (B) Monotonicity (C) Irrevocability (D) None of the given options

11. In best first search, which of the following is true:
 - (A) A set of sorted paths is added to the queue
 - (B) Entire queue is sorted each time a new path is added
 - (C) Chronological backtracking is done to switch to promising paths
 - (D) All the nodes with a specified heuristic value(s) are explored first
12. _____ is a conceptual representation, while _____ is a concrete one.
 - (A) Search Tree, State Space
 - (B) State Space, Search Tree
13. Consider a search tree with depth $d = 3$ and branching factor $b = 2$ the number of nodes explored by depth first iterative deepening search in worst case would be:
 - (A) 8
 - (B) 15
 - (C) 26
 - (D) 22
14. For certain problems, e.g. fractional knapsack problem, hill climbing is always guaranteed to give optimal solution.
 - (A) True
 - (B) False
15. Heuristic evaluation function for the uniform cost search may be characterized as:
 - (A) $f(n) = g(n) + h(n)$; where $h(n) = 0$
 - (B) $f(n) = g(n) + h(n)$; where both $g(n)$ and $h(n)$ are non-negative
 - (C) $f(n) = g(n) + h(n)$
 - (D) None of the given options
16. Admissibility is a stricter requirement than monotonicity.
 - (A) True
 - (B) False
17. TSP optimization problem is an NP hard problem.
 - (A) True
 - (B) False
18. A larger tournament size would result in a more diverse offspring population.
 - (A) True
 - (B) False
19. Genetic algorithm can be characterized as a powerful variant of:
 - (A) Uniform cost search
 - (B) Breadth first search
 - (C) Hill climbing search
 - (D) None of the given options
20. Mutation helps ensure that _____ is present in the offspring population.
 - (A) Stagnation
 - (B) Diversity
 - (C) Inversion
 - (D) None of the given options

PART-2 (50 Marks)

Note: Attempt all questions.

Q.1. [1+2+4+3]

A farmer wants to move himself, a fox, a goose and some grain across a river. Unfortunately, his boat is so tiny that he can take only one of his possessions across the river on any trip and he is the one who could operate the boat. Worse yet, an unattended fox will eat a goose, and an unattended goose will eat grain! So, the farmer must not leave the fox alone with the goose or the goose alone with the grain. If the farmer and his possessions are all on the left bank of the river, what sequence of moves must the farmer follow to get to the right bank?

Consider representing the above problem using the state space representation as follows: Using the 4-tuple representation for a problem state as $\langle a_1, a_2, a_3, a_4 \rangle$, where the indices 1 through 4 represent the farmer, the fox, the goose, and the grain, respectively, and $a_i \in \{L, R\}$, where L and R designate left and right banks of the river, do the following.

- What would the start and goal states look like?
- Give the states that must be avoided given the constraints of the problem.
- Draw the state space graph showing only the permissible moves.
- Give the sequence of moves to reach the goal state. Use the tuple notation for a state and arrows to show transition between the states.

Q.2. [4+6]

- Differentiate between informedness and admissibility of a heuristic.
- Show that monotonicity implies admissibility.

Q.3. [2+4+4]

Consider the pseudo code for the best first search algorithm and answer the following questions.

```
function best_first_search;
begin
    open := [Start]
    closed := []
    while open <> [] do
        begin
            % initialize
            % states remain
            remove the leftmost state from open, call it X
            if X = goal then return the path from Start to X
            else
                begin
                    generate children of X
                    for each child of X do
                        case
                            the child is not on open or closed:
                                begin
                                    assign the child a heuristic value
```

```

        add the child to open
    end
    the child is already on open:
        if the child was reached by a shorter path
            then give the state on open the shorter path
        the child is already on closed:
        begin
            if the child was reached by a shorter path then
                begin
                    remove the state from closed
                    add the child to open
                end
            end
        end
        put X on closed
        % case statements end
    re-order states on open by heuristic merit % (best leftmost)
end
end
return FAIL % open is empty
end

```

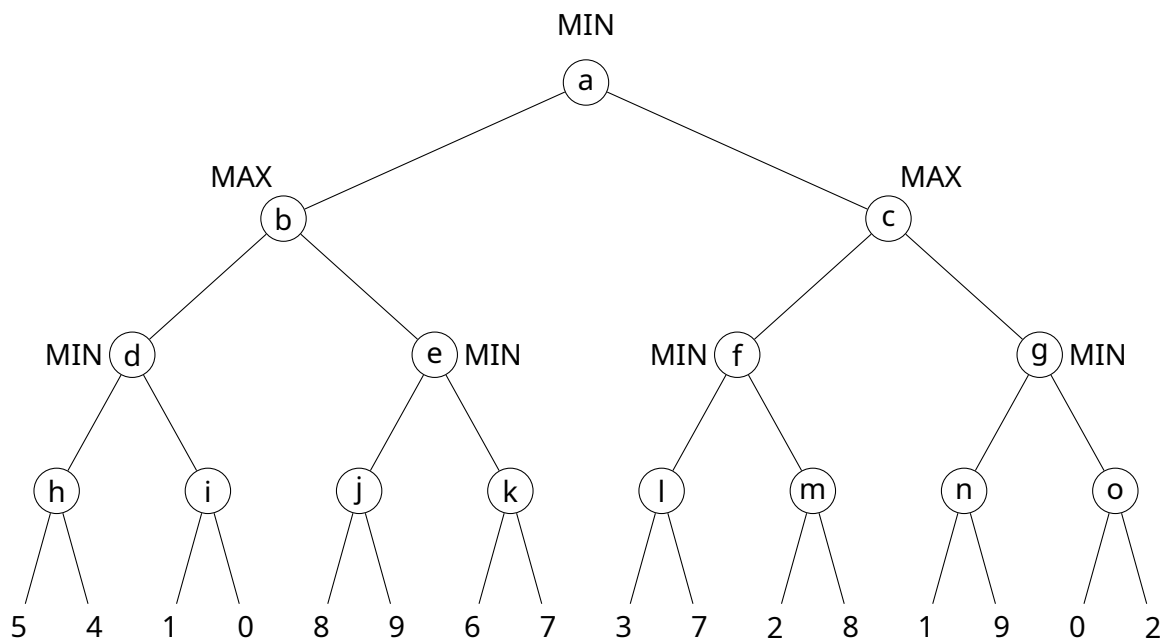
- (a) Does the pseudo code above work fine? Give your answer as a "Yes" or "No".
- (b) Change the pseudo code if you answer "No" in (a) above. Note: Highlight the section you think is not correct and give the modified statement(s) below.
- (c) Modify the given pseudo code if the heuristic you use is consistent (monotone).

Q.4.

(a) How does Alpha-Beta pruning help improve the performance of the Minimax algorithm? Give the total game nodes examined by the Alpha-Beta search in the best as well as the worst-case scenario.

(b) For the search tree shown below it is MIN's turn. You are required to compute the final values of the pair (α, β) for all non-leaf nodes generated by the algorithm and fill them in the given table. What move does the MIN select?

Note: Do the conventional left to right depth-wise traversal. You must NOT evaluate (α, β) at the leaf nodes, which implies that the propagation of (α, β) pair must be stopped one level above the ply depth!



Node	Final (α, β)
a	
b	
c	
d	
e	
f	
g	
h	
i	
j	
k	
l	
m	
n	
o	

Q.5. [2+6+2]

- (a) Consider bit-level representation for traveling salesperson problem involving 'n' cities. Describe one issue each in designing the crossover and mutation operators for this representation.
- (b) For the following candidate solutions, apply order-crossover as discussed in the class to create two new solutions. You must take the cut points as follows: first between city index 3 and city index 4, second between city index 7 and city index 8.

Note: The candidate solutions shown below do not include the starting city. The list indexing starts from 1, NOT 0.

$$P_1 = (1, 9, 2 \mid 4, 6, 5 \mid 7, 8, 3)$$

$$P_2 = (4, 5, 9 \mid 1, 8, 7 \mid 6, 2, 3)$$

- (c) How does tournament selection work in genetic algorithm? What happens if the tournament size is too large?